Fractional Gaussian Fields on the Vicsek Fractal

Master's Thesis Defence

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Vicsek Fractal

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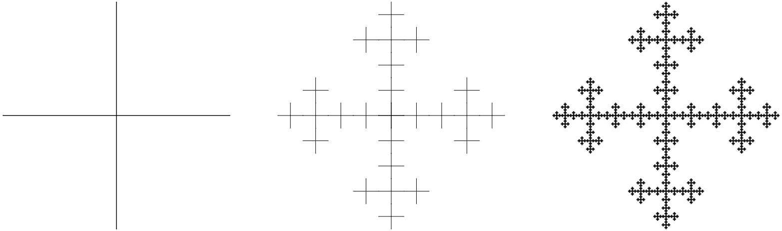

Figure: Vicsek graph, G_0, G_2 and G_5.

Let $x, y \in K$ then $\{x\} = x_{i_1,i_2,\ldots}$ and $\{y\} = y_{j_1,j_2,\ldots}$. If $x \underset{m}{\sim} y$ they must share an a α such $i_m = j_m$ for $m \leq \alpha$.

Lemma 2.4

Let $u : V^* \to \mathbb{R}$ be a harmonic function on the Vicsek set . If $v(x) = v(y) = 0$ and $y \in V_{m,x}$, then $v(\omega) = 0$ for all $\omega \in T_x$ where $T_x = \left(\bigcup_{\mathcal{I}_{i_1,\ldots,i_\alpha}} \mathcal{S}_{i_1,i_2,\ldots}(K)\right)$.

Proof.

Clear $v(\omega) = 0$ for $\omega \in T_x \bigcap V^*$ since v is harmonic and depends on the values of its neighbours.
Now let $\omega \in T_x \setminus V^*$ then $T_x \setminus V^*$ is dense in T_x and the same arguments follow. □